Scores and Calibration Intuition

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

Teach one probability event consistently.

Supply all lab records, declared endpoints and separate fixture counts; preserve the distinct confidence example as optional review.

Keep one habit: Require denominators, gap directions and honest small-sample/population limits.

MAIN LESSON · SECTION 1

Prepare the connected experiment

> Keep prior exposure, shared practice and independent evidence distinct.

Goal and main teaching plan

Primary target: Interpret **calibration** using groups of forecasts

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Sections 1–6; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Sections 7–10; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Materials and prerequisite

Teach two 45-minute sessions. This week focuses on **calibration** as a group property. Model the worked example first, collect the new W 4 case independently before feedback, and use the fresh retry to check transfer.

Award credit for traceable reasoning and bounded claims, including justified uncertainty.

Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.

No. Its interpretation must be established; a threshold can act on a **score** without making it a calibrated probability.

Printed student lesson and workbook

Paper, pencils and a calculator for fractions

The fictional examples and a blank **evidence** table

Optional Python 3 editor; no external packages or AI accounts required

Opening retrieval

Without opening the old lesson, explain one key idea from Train a Classifier. Give an example and a mistake that the idea helps you avoid.

Look for: Connect **training**-only scaling and neighbour **prediction** in one pipeline

Reference example, not the only acceptable wording: fit learns **training** minima and ranges and stores scaled **training** rows with IDs and labels. predict takes a raw feature pair and the fitted state.

Evaluation compares a **prediction** with its reference label afterward. Validation references belong only in evaluation, not in **fitting** or the neighbour pool.

Without opening the old lesson, explain one key idea from Decision Trees. Give an example and a mistake that the idea helps you avoid.

Look for: Fit a root question and leaf **prediction**s from **training** examples

Reference example, not the only acceptable wording: The root asks near, with branches 0 and 1. Leaves predict clear for 0 and blocked for 1. Depth is one.

Training labels determined both leaf majorities and the error comparison that selected near rather than tall.

Without opening the old lesson, explain one key idea from Leakage. Give an example and a mistake that the idea helps you avoid.

Look for: Choose a split unit that matches the intended **prediction** claim

Reference example, not the only acceptable wording: **Training** fits the model; validation compares development choices; test evaluates after choices are fixed.

Repeatedly selecting by the test **score** uses that set for development, so it is no longer an untouched final evaluation.

End goal

By the end, I can explain **score**s and **calibration** intuition using a traced example and a stated limitation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Train a Classifier

Decision Trees

Leakage

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Twenty cases receive **score** .7 and fourteen are correct. Explain what the observed rate suggests and what it does not establish about another **score** group.

Concept to explain

A model **score** can rank **prediction**s without matching the real frequency of correct outcomes.

Term	Meaning
score	model value used for ranking or confidence
calibration	scores align with observed frequencies

</details>

Term	Meaning
bin	group of predictions with similar scores
reliability	agreement between score and outcome rate

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach positive-class calibration before any confidence analogy

> Keep the event and reference outcome unchanged within a comparison.

Score-event contract and group interpretation

Current score meaning	Correct comparison
p(positive), here p(spotted under a toy label rule)	Compare mean predicted p with fraction of reference-positive cases in that score group.

Current score meaning	Correct comparison
p(chosen label is correct), a separate confidence-calibration task	Compare with fraction of selected predictions that were correct; do not substitute this outcome into the positive-class example.
Arbitrary model score or neighbour vote share	Do not call it a probability without a justified interpretation and evaluation.
Evidence limit	Agreement in a small group is an observation, not a guarantee for an individual, another score bin or a new population.

If many comparable cases receive probability .8, a calibrated model should produce positives in roughly 80 percent of those cases over adequate data.

It does not mean every case is 80 percent correct or that exactly eight of each next ten must be positive. **Calibration** is assessed across groups.

Use fictional classroom data throughout this investigation. Keep the task, units, reference labels and data split visible as you reason.

Earlier weeks supplied the model-building **process**; this week examines **calibration** as a group property.

Original two-group calculation

In a fictional held-out check, ten cases all have predicted probability .2 and two are positive. Another ten all have probability .8 and six are positive.

Average **prediction**s are .2 and .8; observed frequencies are 2/10=.2 and 6/10=.6. The second group is overconfident for positives in this sample.

Trace the calculation on paper before running this optional Python check. Use the stated fictional **input**s and compare intermediate values. The code checks this example; your independent workbook case uses different **evidence**.

Term	Meaning
Calibration	Agreement between predicted probabilities and outcome frequencies across comparable groups
Bin	A declared interval grouping predictions

Term	Meaning
Observed frequency	Positive outcomes divided by cases in a group
Reliability diagram	A plot comparing group average prediction with observed frequency

Calibration as a group property at a glance

PART	EVIDENCE OR ACTION	INTERPRETATION
Low probability group	Average .2	Observed 2 of 10
High probability group	Average .8	Observed 6 of 10
Compare groups	Prediction versus frequency	Keep group sizes visible

Plot average prediction on the horizontal axis and observed frequency on the vertical axis. The two points are (.2,.2) and (.8,.6). Agreement lies along the diagonal. Use the average score within a bin, not automatically the bin midpoint. Small groups make frequencies uncertain.

Which worked group overstates positive frequency and by how much? Explain the steps or evidence behind your response.

Separate pictured groups and direction

Treat **calibration** as a comparison across groups of probability **score**s.

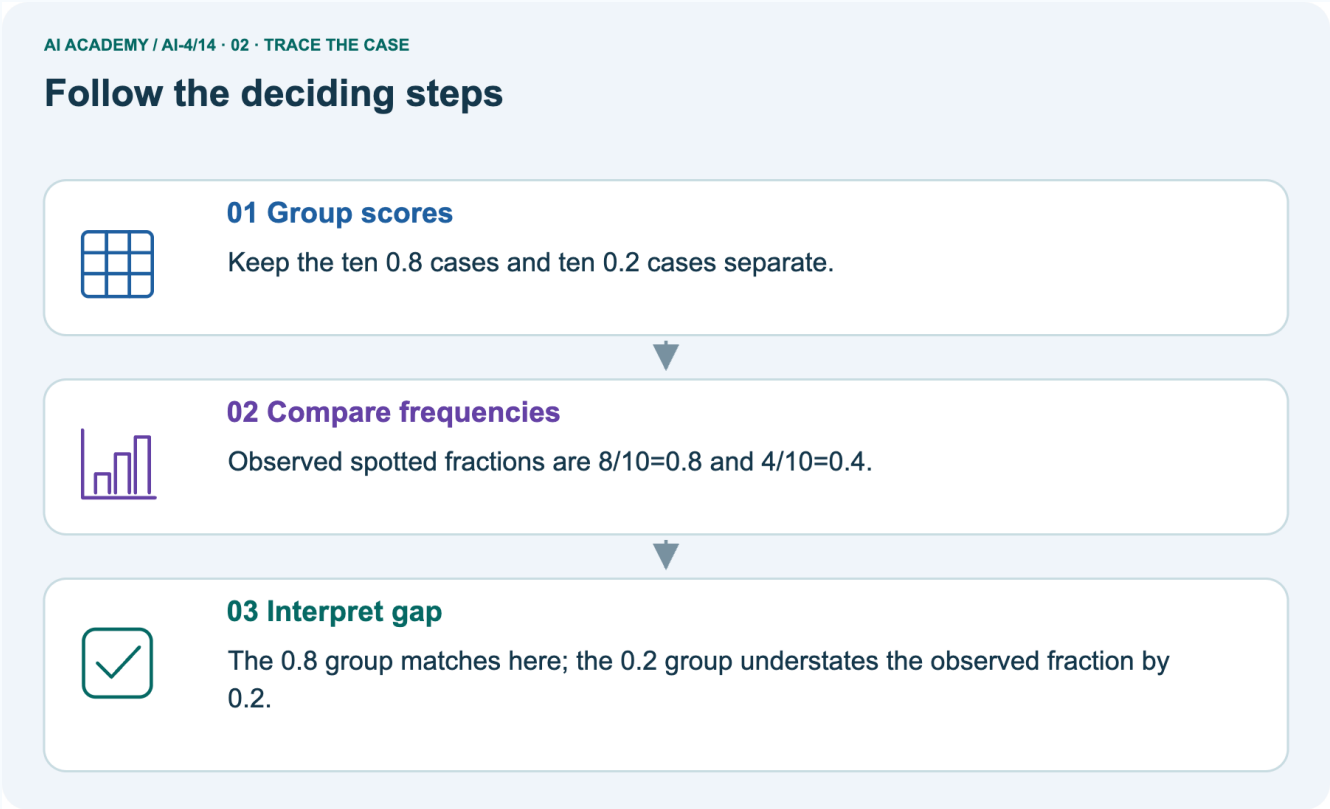
Toy classifier **output**s p(spotted)=0.8 on ten cases; eight are actually spotted. Another ten have p=0.2; four are spotted.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The main calculation and picture are different fixtures. Main low group: p=0.2,2positives/10,point(0.2,0.2); main high: p=0.8,6/10,point(0.8,0.6),overstates by0.2 or20percentage points. Picture high: p=0.8,8/10; picture low: p=0.2,4/10,understates by0.2. For a reliability diagram x is mean predicted positive probability and y observed positive fraction; compare with y=x.

Neither example tells whether every chosen class label is correct. Ten cases per group remain noisy evidence.

Trace frequencies and changed outcome count



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Group scores	Keep the ten 0.8 cases and ten 0.2 cases separate.
2. Compare frequencies	Observed spotted fractions are 8/10=0.8 and 4/10=0.4.
3. Interpret gap	The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Read the picture in order. The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Ten cases per group provide noisy **evidence**; the exercise does not establish **calibration** on every **score** or setting.

Change: The 0.2 group has two spotted cases instead of four.

Prediction: Its observed fraction becomes 0.2 and matches the supplied **score** in this group.

Reason: **Calibration** compares stated probabilities with group outcome frequencies.

▼ Additional chosen-label-correctness example

UNDER THE HOOD For this example, each **score** estimates the probability that the selected label is correct. Ten **prediction**s have **score**s near 0.8; six are correct, so observed correctness is 6/10=**60%**.

If instead **score**s mean probability of the spotted class, compare them with the fraction of reference-spotted leaves.

STEP 1 Each **score** estimates the probability that the selected label is correct. Ten **prediction**s have such **score**s near 0.8.

STEP 2 Only six are correct, so observed correctness is 6 ÷ 10 = **60%**.

STEP 3 The **score** ranking may still help, but 0.8 should not be stated as an **80%** guarantee.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Trace the information, fitted behavior, and evaluation proof separately.

QUESTION Bin **prediction score**s and compare average **score** with observed correctness.

MODEL IDEA A model **score** can rank **prediction**s without matching the real frequency of correct outcomes.

PROOF NEEDED Use an unseen case, visible calculation, error analysis, and a stated limitation.

Draw a fresh data → model → **prediction** → evaluation trace:

Predict first. Then check each step. A correct answer without visible **evidence** cannot be audited.

STEP 4 Use bins across many cases and design an uncertain zone requiring human review.

CHECK Verify the denominator, unit, split, class definition, and data version before interpreting a **score**.

Recalculate, retrace, or restate the **evidence** independently:

Concrete teaching case

Ten cases receive probability .8 for spotted; eight are spotted.

Interpret **calibration** and its limits.

Exact explanation to model

8/10 matches .8 in this tiny bin. It illustrates **calibration**, not proof of global **calibration** or individual certainty; test more bins and data.

Explain the added worked example

Key terms

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

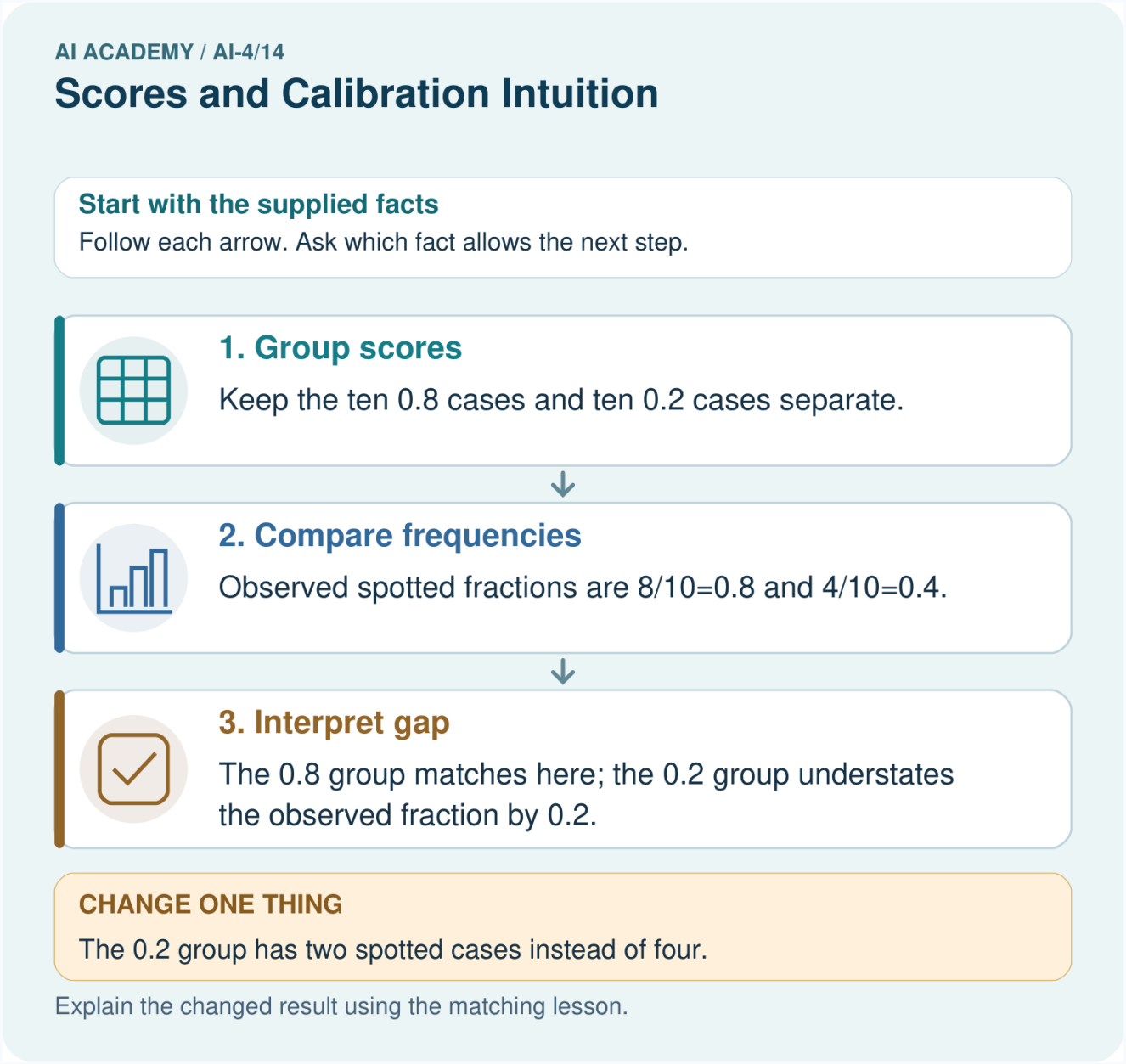
These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Scores and Calibration Intuition

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Guide the calculation, then remove help

> Collect each learner's prediction before showing its key.

Current completion task and checked key

Completion task: Which worked group overstates positive frequency and by how much?

Teacher check: Teacher explanation: The .8 group exceeds observed .6 by .2, or 20 percentage points in this sample.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Calibration** as a group property. Proposed explanation: "A final answer is enough". Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: Ask for the relevant counts, units, procedure or saved record. For this week: 2/10=.2 and 6/10=.6.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial question and changed-case reasoning

Does a calibrated 0.8 **score** mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

No. It describes a group proportion: here eight of ten are spotted, and two are not. It is not an individual guarantee. Changed-case check: Its observed fraction becomes 0.2 and matches the supplied **score** in this group.

Calibration compares stated probabilities with group outcome frequencies.

Changed-case reasoning: Its observed fraction becomes 0.2 and matches the supplied **score** in this group. **Calibration** compares stated probabilities with group outcome frequencies.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Keep each learner explaining

Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Compact alternative: this two-session sequence covers selected core tasks. Use the three-session weekly plan for the full lesson, independent transfer, feedback and project work; allow extra time for extensions.

Demonstrate and guide practice

Board sequence

Write the given facts: Ten cases receive probability .8 for spotted; eight are spotted.

Write the question to explain: Interpret **calibration** and its limits.

Reveal after **prediction**s: 8/10 matches .8 in this tiny bin. It illustrates **calibration**, not proof of global **calibration** or individual certainty; test more bins and data.

Use the earlier transfer question as additional practice. Reserve the fresh two-group positive-probability interpretation task for the first individual attempt and record support used.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

Guided response: The .8 group exceeds observed .6 by .2, or 20 percentage points in this sample. Exit response: The reported improvement reuses the answers used for adjustment; evaluate the frozen adjustment on separate data.

Case-specific teaching sequence

Stage · Teacher action

> **Stage:** Session 1 start to 7 minutes
>
> **Teacher action:** Use the prerequisite question: Does a score necessarily represent a probability? Check the explanation before proceeding.

> **Stage:** 7 to 18 minutes

Teacher action: Read the opening and model the worked evidence. Ask students to identify calibration in context.

Stage: 18 to 32 minutes

Teacher action: Complete W 1 and W 2 with support: compute frequencies; locate the discrepancy. Require the evidence, not only the final response.

Stage: 32 to 45 minutes

Teacher action: Read the visual row by row. Collect the guided explanation and compare reasoning in pairs. Trace the Python check after the paper calculation.

Stage: Session 2 start to 8 minutes

Teacher action: Retrieve the distinction between calibration and bin. Invite a counterexample to an overconfident claim.

Stage: 8 to 20 minutes

Teacher action: Discuss the application and complete W 3: Why can constant .5 predictions look calibrated but rank poorly?

Stage: 20 to 35 minutes

Teacher action: Collect W4 independently before giving hints. Assess these outcomes: Interpret calibration using groups of forecasts; Compare average predicted probability with observed frequency. Record any support used.

Stage: 35 to 45 minutes

Teacher action: Use W 5 and the exit response to assess limits. Give targeted feedback, then the fresh retry. Save evidence of independent understanding.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Keep predictions, observed or supplied outcomes and unmeasured cases separate.

Supplied inputs and checked guidance

Thirty-card key:10cases per bin; positive counts2,7,6;mean scores0.2,0.5,0.8;observed0.2,0.7,0.6. Middle understates0.2,high overstates0.2. p=0.35belongs to middle; empty group unavailable,unknown outcome notnegative; changed-population0.8/3of10has gap0.5but very little evidence. Independent20at0.3with8positives→(0.3,0.4),understates0.1;10at0.7with7positives→(0.7,0.7), matches here.

No individual or global guarantee and no calibrator fitted.

Predict, investigate and record

1. 1. Write the hypothesis and expected **evidence** before running.
2. 2. **Reliability** Lab: sort the thirty supplied p(positive) records into the declared bins; compare average **score** with observed positive frequency and retain each denominator.
3. 3. Run one normal, one boundary, one missing, and one shifted case.
4. 4. Calculate the metric from raw counts or row-level errors.

5. 5. Interpret what changed, what did not, and what remains unsafe or unknown.

Keep the experiment ledger

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the current misconceptions

A final answer is enough: Ask for the relevant counts, units, procedure or saved record. For this week: 2/10=.2 and 6/10=.6.

A stronger claim makes a stronger project: The reported improvement reuses the answers used for adjustment; evaluate the frozen adjustment on separate data.

The new case can be copied from the example: The independent task changes the **evidence**. Require a fresh explanation: Points (.3,.4) and (.7,.7). The first understates **observed frequency** by .1; the second agrees on this sample.

Group sizes still limit claims.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Keep the test conditions fixed, record every result, and change only one planned factor.

Make the lab runnable

Use the supplied complete investigation for shared practice. Reserve the fresh two-group positive-probability interpretation task for the individual first attempt before hints or feedback; earlier transfer questions remain additional practice.

Diagnose and reteach

MISCONCEPTION TO REPAIR A **score** of 0.9 means the **prediction** has exactly a 90% chance of being correct.

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the selected experiment transfer

> Keep this key until after a first attempt; record any help and prior exposure.

Selected independent task and checked key

Selected case: Twenty cases all predict .3 with eight positives; ten all predict .7 with seven positives. Give plot points and interpret.

Teacher explanation: Points (.3,.4) and (.7,.7). The first understates **observed frequency** by .1; the second agrees on this sample. Group sizes still limit claims.

Judge trace, scope and support

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Compute frequencies

Give the observed frequencies for the two worked groups.

Teacher explanation: 2/10=.2 and 6/10=.6.

Use each group's observed positive count over its observed total: 2/10 = 0.2 and 6/10 = 0.6. These are sample frequencies for the worked groups.

Do not turn either small-group **observation** into a guarantee about the next individual or a proof of **calibration**.

W2 Locate the discrepancy

Which worked group overstates positive frequency and by how much?

Teacher explanation: The .8 group exceeds observed .6 by .2, or 20 percentage points in this sample.

W3 Separate ranking

Why can constant .5 **prediction**s look calibrated but rank poorly?

Teacher explanation: When half the cases are positive, the overall fraction agrees, but every case has the same **score** and no ranking is provided.

W5 Protect evaluation

What is wrong with calibrating and reporting success on the same final-test records?

Teacher explanation: The reported improvement reuses the answers used for adjustment; evaluate the frozen adjustment on separate data.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

8/10 matches .8 in this tiny bin. It illustrates **calibration**, not proof of global **calibration** or individual certainty; test more bins and data.

Additional transfer practice question

Among 20 **prediction**s with **score** 0.8, 12 are correct. What is observed accuracy in this group and how does it compare with 0.8?

Reasoned practice answer

Observed accuracy is 12/20 = **60%**, below the **80% score**. This small group suggests overconfidence, but more data is needed to assess **calibration** reliably.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A model **score** of 0.8 guarantees that this particular **prediction** is correct.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

14/20=.7 matches the stated **score** for this small group, illustrating **calibration** if **score**s mean estimated correctness probability. It does not establish global **calibration**, stability or certainty for one case.

Examine more data and other bins with a clear **score** meaning.

Explanations for the additional workbook examples

W1 Compute frequencies

W2 Locate the discrepancy

W3 Separate ranking

W4 Apply independently

Twenty cases all predict .3 with eight positives; ten all predict .7 with seven positives. Give plot points and interpret.

Teacher explanation: Points (.3,.4) and (.7,.7). The first understates **observed frequency** by .1; the second agrees on this sample. Group sizes still limit claims.

W5 Protect evaluation

Assess independent transfer

Selected case answer

Feedback that the learner uses

Connect, support and revisit the experiment

> Retain failures and assess a fresh parallel case after help.

Project evidence

Add an uncertainty band to the Plant Classifier and define what the interface says when **score**s fall inside it.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Support and selected reassessment

Use a tiny labelled dataset, explicit units and a named positive class. Stretch: require a baseline, held-out evaluation and subgroup or shifted-condition analysis with denominators.

Provide a blank **evidence** table and allow a verbal explanation before writing. Highlight the given conditions without supplying the W 4 conclusion. Use **calibration** and bin in full sentences.

Extension: alter one condition in the worked example and explain which conclusion changes and which remains supported.

For independent W 4, require: Points (.3,.4) and (.7,.7). The first understates **observed frequency** by .1; the second agrees on this sample. Group sizes still limit claims.

Fresh retry: five .6 **prediction**s with two positives give (.6,.4), an overstatement of .2 in a very small group. Require a small-sample limitation. Mark each criterion as independent, supported or not yet demonstrated.

Reassess the reasoning after feedback; do not replace the first attempt silently.

Delayed return

Suggested return points in this level: ai-4/15, ai-4/17, ai-4/20. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge and reflection

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 15: Confusion Matrix. Retrieve today's **evidence** and identify what carries forward and what is new.

Next is Week 15: Confusion Matrix. Carry forward the positive-class definition, score meaning and ID-linked references/predictions; error analysis is developed later. Primary teacher reference: https://scikit-learn.org/stable/modules/calibration.html .

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Responsible Plant Classifier

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route without repeating the whole cycle.

Planning and assessment reference

Teaching target: Interpret **calibration** using groups of forecasts

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
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> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use “Run the investigation”, “Connect your project”, “Show what you know”, “Study a complete case” and “Try a new case independently”. Give extra time for the full task set. If this week has a coding lab, use its named section; section numbers can change when extra explanations are included.

Before class, solve the supported practice: Which worked group overstates positive frequency and by how much?

Reasoned practice answer: Teacher explanation: The .8 group exceeds observed .6 by .2, or 20 percentage points in this sample.

Check every learner before discussion: Use the worked case for **Calibration** as a group property. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: Ask for the relevant counts, units, procedure or saved record. For this week: 2/10=.2 and 6/10=.6.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: The 0.2 group has two spotted cases instead of four. Require a **prediction** and a reason before comparing. Expected result: Its observed fraction becomes 0.2 and matches the supplied **score** in this group.

Reason: **Calibration** compares stated probabilities with group outcome frequencies.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback and evidence records**

REFERENCE · SECTION 8

Reference · Feedback and evidence records

> Keep supported rehearsal separate from independent performance.

Record and repair

Use the two formative questions as supported practice, not as independent mastery **score**s. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

▼ **Earlier wording, alternatives and picture transcripts**

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

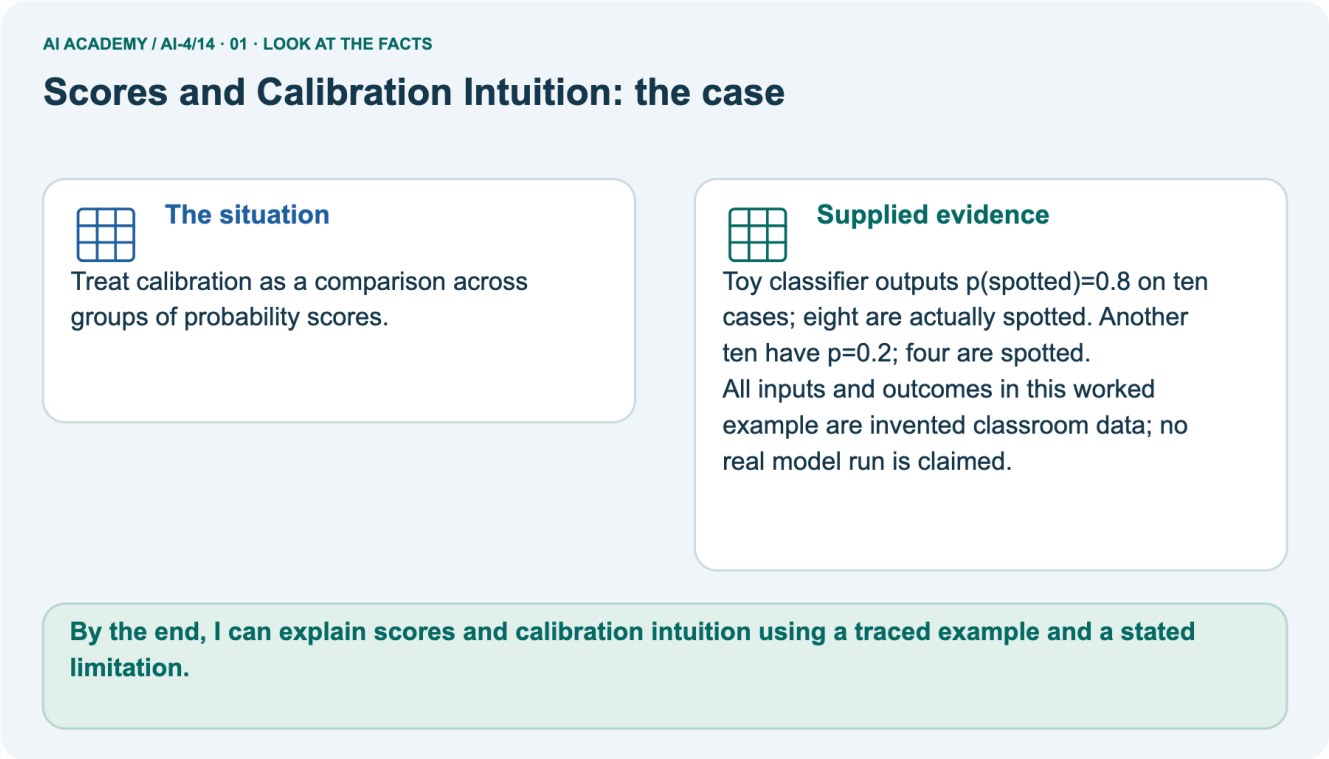
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a specific reasoning move.

01. Scores and Calibration Intuition: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

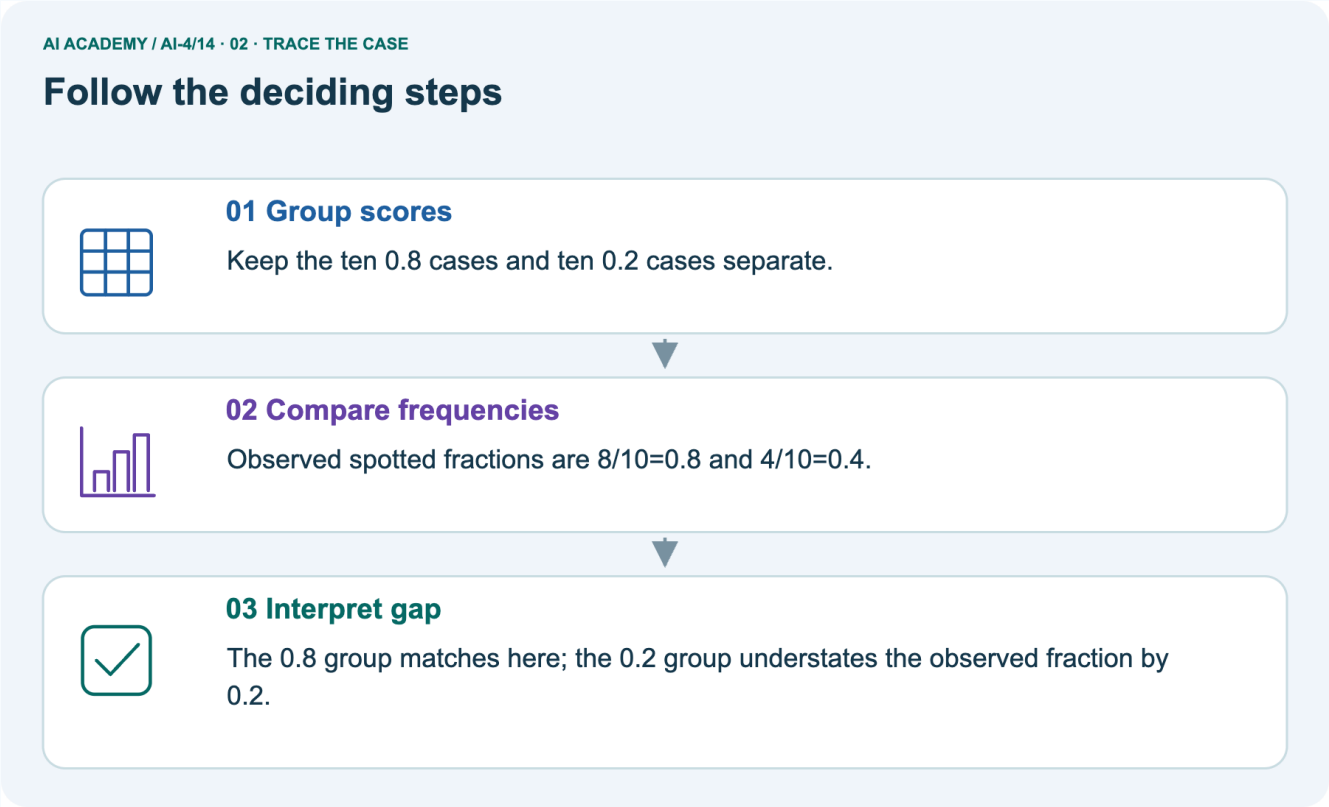
Treat **calibration** as a comparison across groups of probability **score**s. Supplied facts: Toy classifier **output**s p(spotted)=0.8 on ten cases; eight are actually spotted. Another ten have p=0.2; four are spotted.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Group **score**s: Keep the ten 0.8 cases and ten 0.2 cases separate. → Compare frequencies: Observed spotted fractions are 8/10=0.8 and 4/10=0.4. → Interpret gap: The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Four-step worked evidence for Week 14: Scores and Calibration

Intuition

AI ACADEMY / AI-4/14 · 03 · INSPECT THE LESSON MODEL

Four-step worked evidence for Week 14: Scores and Calibration Intuition

Step	Evidence and reasoning
1	Each score estimates the probability that the selected label is correct. Ten predictions have such scores near 0.8.
2	Only six are correct, so observed correctness is 6 ÷ 10 = 60%.
3	The score ranking may still help, but 0.8 should not be stated as an 80% guarantee.
4	Use bins across many cases and design an uncertain zone requiring human review.

Trace these corrected steps. Keep the stated metric, assumptions and interpretation when you change the case.

Follow the labelled picture with the matching explanation. Open the vector diagram

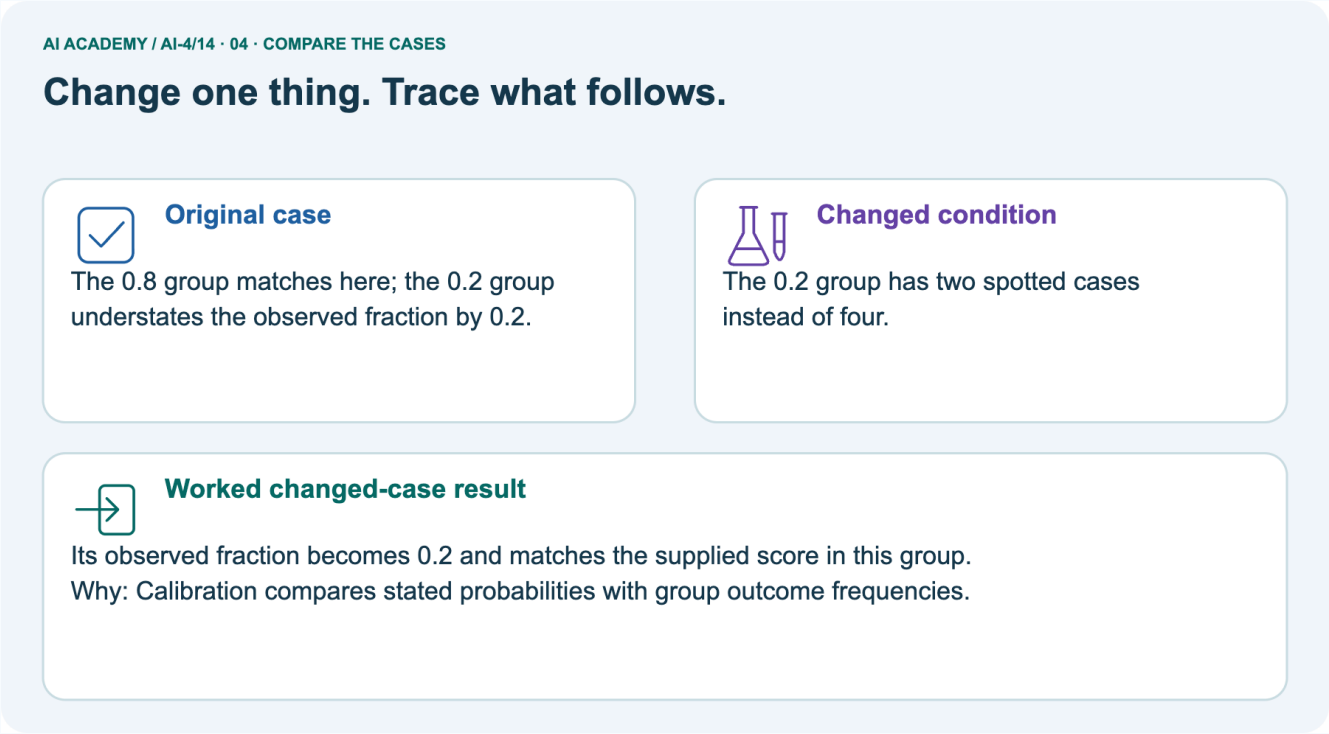
Four-step worked **evidence** for Week 14: **Score**s and **Calibration** Intuition. 1 | Each **score** estimates the probability that the selected label is correct.

Ten **prediction**s have such **score**s near 0.8.. 2 | Only six are correct, so observed correctness is 6 ÷ 10 = **60%**.. 3 | The **score** ranking may still help, but 0.8 should not be stated as an **80%** guarantee.. 4 | Use bins across many cases and design an uncertain zone requiring human review.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. Changed condition: The 0.2 group has two spotted cases instead of four.

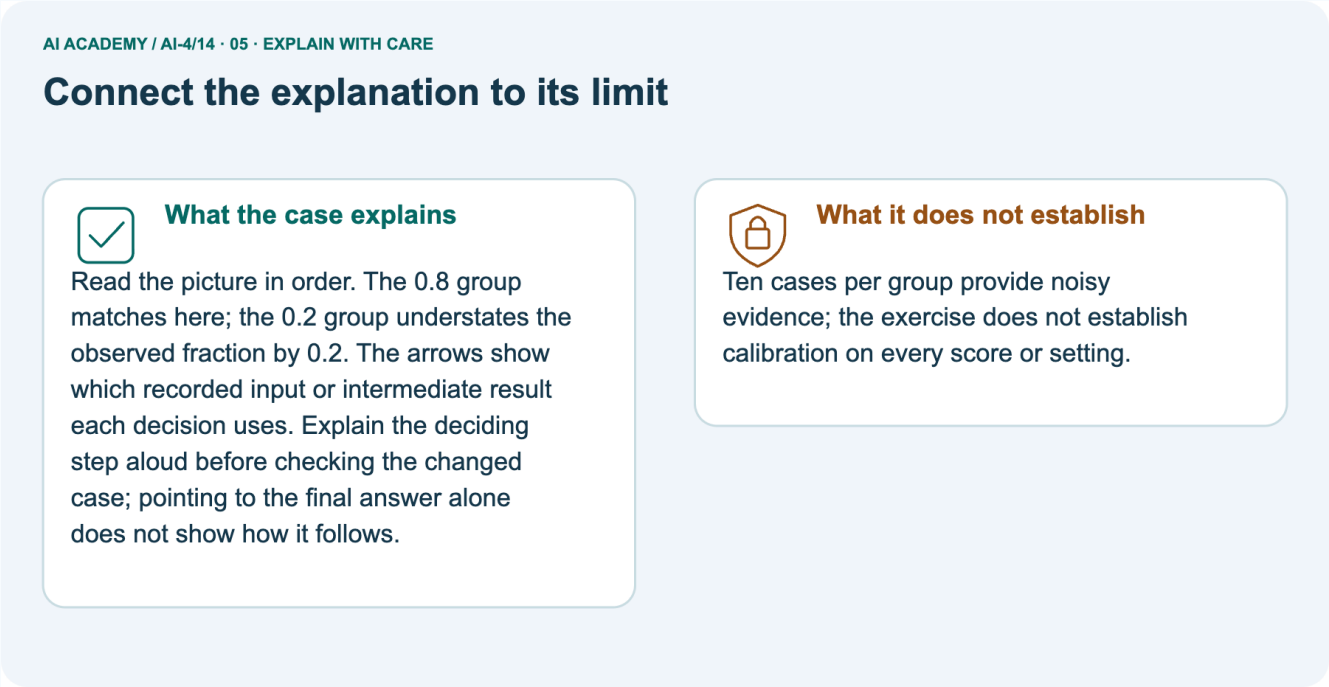
Expected result: Its observed fraction becomes 0.2 and matches the supplied **score** in this group. Reason: **Calibration** compares stated probabilities with group outcome frequencies.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: **Calibration** compares stated probabilities with group outcome frequencies.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. The 0.8 group matches here; the 0.2 group understates the observed fraction by 0.2. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: Ten cases per group provide noisy **evidence**; the exercise does not establish **calibration** on every **score** or setting.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Ten cases per group provide noisy **evidence**; the exercise does not establish **calibration** on every **score** or setting.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Does a calibrated 0.8 **score** mean every such leaf is spotted? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: No. It describes a group proportion: here eight of ten are spotted, and two are not. It is not an individual guarantee.

Changed-case check: Its observed fraction becomes 0.2 and matches the supplied **score** in this group. **Calibration** compares stated probabilities with group outcome frequencies.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Scores and Calibration Intuition: the case

02. Follow the deciding steps

03. Four-step worked evidence for Week 14: Scores and Calibration Intuition

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.